

Haiyi Xing

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EDUCATION

University of California, San Diego - San Diego, CA | Sep 2024 – June 2026

Master of Science in Computer Science and Engineering | GPA: 3.8/4.0

University of California, Irvine - Irvine, CA | Sep 2020 – June 2024

Bachelor of Science in Computer Science and Engineering | Minor in Mathematics | GPA: 3.84/4.0

TECHNICAL SKILLS

Model Architectures: Transformers (BERT, GPT, Llama), MoE, CNNs, RNN/LSTM

LLM & Agent: Hermes, Openclaw, LoRA, P-Tuning v2, Reinforcement Learning, Prompt Engineering, LangChain

Deep-Learning Frameworks: PyTorch, TensorFlow

Programming Languages: Python, C++17, Java, Go

Data & Databases: MySQL, Milvus

Workflow & Tooling: Git, Docker, Linux, AWS

PROFESSIONAL EXPERIENCE

Real-Time Neuro-Feedback Pipeline — Alzheimer's Edge Project - 02/2025 – 04/2026

- Re-engineered MATLAB EEG-feedback suite into C++17, reducing end-to-end latency by 86% (270ms → 38ms).
- Deployed browser-based demo via WebAssembly, streaming 256Hz multi-channel EEG at 60 fps under 5MB heap.
- Trained CNN-UNet for artifact detection, boosting signal SNR by +18dB (F1: 0.92) and reducing reconstruction RMSE to 0.06.

Yuanjing LLM Instruction-Prompting Intern, China Unicom - 07/2024 – 09/2024

- Engineered and A/B-tested 150+ specialized prompts using few-shot in-context learning, increasing response relevance by 17% and decreasing hallucinations by 12%.
- Spearheaded a large-scale data initiative by cleaning and labeling ~2.3M telecom QA pairs, reducing model perplexity by 15% through optimized retraining pipelines.
- Orchestrated a strategic Python/LangChain evaluation harness for enterprise-grade regression testing, now the standard for performance tracking across 3 cross-functional teams.

Software Development Engineer Intern, Lenovo - 08/2023 – 10/2023

- Architected a complex multi-stage AI support system leveraging RAG (BERT) and synthesis (ChatGLM), automating resolution for 80% of device-specific technical inquiries.
- Deployed a high-precision intent-recognition engine using ensemble methods (LR + GBT), boosting F1 scores from 78% to 92% to optimize enterprise task routing.
- Standardized machine learning production workflows by leading technical workshops for 30+ engineers, establishing best practices for scalable model deployment.

NAPLEX LLM Benchmark, Pierre Baldi Lab - 09/2023 – 12/2023

- Pioneered a rigorous benchmarking framework to evaluate frontier LLMs (GPT-4, Llama-2-70B) on the NAPLEX medical exam, uncovering critical scaling laws in specialized domains.
- Validated the efficacy of parameter scaling through statistical analysis, demonstrating a 15% accuracy surge during the 7B to 70B parameter transition.
- Co-authored a bioRxiv preprint on dataset curation and zero-shot optimization, contributing novel methodologies for evaluating AI performance in healthcare.

Genomic-Sequence Health-Risk Modeling, Jing Zhang Lab - 03/2023 – 10/2023

- Innovated genomic health-risk modeling by replacing baseline CNNs with hybrid ResNet-Transformer architectures, cutting inference latency by 30% while preserving accuracy.
- Optimized Hyena-based long-sequence models for high-dimensional biological data, achieving a 40% reduction in VRAM usage and doubling batch processing throughput.

PROJECTS

Latent Reasoning Steering for LLMs

- Investigated latent-space reasoning by implementing steering vectors to intervene in model hidden states, enabling dynamic control over reasoning depth.
- Analyzed performance trade-offs on AIME math benchmarks through rigorous ablation studies, quantifying the impact of inference-time interventions on pass@1 accuracy.